Yuchen Zhu | 朱雨宸

Generative AI Researcher | Agentic LLMs, Coding Agents, Diffusion Language Models

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Google Scholar | GitHub | LinkedIn

Seeking full-time roles beginning Dec 2026

Professional Summary

Final-year Machine Learning Ph.D. candidate at Georgia Tech building effective and efficient generative AI systems. Research spans agentic LLMs and coding agents, LLM post-training and evaluation, and diffusion language models, from algorithms to large-scale systems. Industry research experience at NVIDIA and Adobe Research; first/co-first author of 9 ICML, ICLR, and NeurIPS papers, including an ICML 2026 Spotlight.

- **Agentic LLMs:** data-centric mid/post-training, reinforcement learning, and model evaluation.
- **Efficient LLM Systems:** full-stack diffusion-LLM training and serving across pretraining, mid-training, post-training, inference, and custom kernel optimization.
- **Multimodal Diffusion Models:** post-training for image, video, and audio diffusion models.

Education

Georgia Institute of Technology — **2023–Expected Dec 2026**
Ph.D. in Machine Learning; Advisors: Molei Tao and Yongxin Chen — GPA: 4.0/4.0

Yale University — **2022–2023**
M.A. in Statistics and Data Science — GPA: 4.0/4.0

New York University — **2018–2022**
B.S. in Mathematics (Honors), *summa cum laude* — GPA: 3.96/4.0

Industry Research Experience

NVIDIA — **May 2026–Present**
Research Scientist Intern — Santa Clara, CA

- Conduct mid- and post-training research for Nemotron-Cascade 3, Nemotron 3.5, and Nemotron 4, targeting agentic software-engineering and cybersecurity tasks.
- Develop an agentic pipeline for automated curation of RL environments for coding-domain tasks. Built 10k+ high quality environment containers for training-data scarce tasks.

Adobe Research — **Jan 2026–May 2026**
Research Scientist Intern — Seattle, WA

- Led research and development of FLARE, converting Qwen3.5 hybrid-attention autoregressive models (2B/4B/9B) into diffusion LLMs in a single ∼10B-token training stage; FLARE-9B retained 95–99% of its source model's performance on representative math and knowledge benchmarks.
- Designed and implemented new Triton kernels for efficient hybrid attention and two-stream training, fusing recurrent-state scheduling and convolution; increased model FLOPs utilization from 13.80% to 24.81% and reduced peak kernel memory from 18.14 GiB to 0.45 GiB.
- Co-developed a unified SGLang stack supporting speculative and parallel decoding; FLARE delivered up to 4.8× the decoding throughput of open diffusion LLM baselines.

Honors & Awards

ICML 2026 Spotlight (top 2.2%)	2026
NeurIPS Scholar Award	2025
Cloud Hub AI for Science Azure Credits Award ($29,570)	2026
David L. Brown Fellowship Award, Georgia Institute of Technology	2025
Dean's Award in Arts and Sciences, NYU Shanghai (1 of 500)	2022
Major Honors in Mathematics, New York University (1 of 30)	2022

Selected Research & Publications

Selected work most relevant to generative AI research. Full publication list on Google Scholar.
* Equal or core contribution.

FLARE: Diffusion for Hybrid Language Model — **Preprint, 2026**

Yuchen Zhu, Jing Shi, Chongjian Ge, et al.

- Developed a systematic AR-to-dLLM conversion framework for Qwen3.5 hybrid-attention models (2B/4B/9B), combining token-balanced two-stream training with custom Triton kernels for recurrent-state scheduling. A single checkpoint supports both speculative and parallel decoding.
- FLARE-9B retained 95–99% of its source model's math/knowledge performance, while FLARE-2B reached 2,087 tokens/s on one A100 at concurrency 8, up to 4.8× the throughput of open dLLM baselines.

Thinking with Anchors: Grounded and Efficient Document Reasoning — **Preprint, 2026**

Sichen Zhu*, Yuchen Zhu*, Wenzhuo Xu, Jason Kuen, Wanrong Zhu, Jing Shi, et al.

- Co-developed ADOPD 2026, a 120K-page corpus spanning 1,000+ document types that enriches visual anchors with human-cleaned captions, semantic tags, and region-grounded reasoning traces for verifiable document understanding.
- Introduced DocCount, a dense document-counting benchmark with polygon-grounded reasoning; GPT-5.5 and Claude Opus 4.6 both scored below 65% exact-count accuracy.

RMA: Context-Orchestrated Research Math Agents — **Preprint, 2026**

Zelin Zhao, Bo Yuan, Yuchen Zhu, Jaemoo Choi, and Yongxin Chen.

- Co-developed a persistent, issue-driven research-math agent that orchestrates bounded context to identify and repair missing proof obligations; achieved a 42.5% solve rate on SOOHAK Challenge Hard versus 12.5% for its standalone Claude Opus 4.8 backbone, and achieved 8/10 expert-approved solutions on both First Proof B1 and B2 under blind review.

Agents' Last Exam — **Preprint, 2026**

Agents' Last Exam Team; contributing author.

- Contributed to ALE, a long-horizon benchmark spanning 55 professional sub-industries, 1,500+ verifiable tasks, and 300+ domain experts; leading agents remain below 1% average full-pass rate on its hardest tier.
- Lead the curation of verifiable computational-mathematics workflows for agent evaluation.

Enhancing Reasoning for Diffusion LLMs via Distribution Matching Policy Optimization — **ICML 2026 Spotlight**

Yuchen Zhu, Wei Guo, Jaemoo Choi, Petr Molodyk, Bo Yuan, Molei Tao, and Yongxin Chen.

- Developed DMPO, an off-policy, forward-only RL method that matches a dLLM to a reward-tilted target distribution without supervised fine-tuning; introduced weight-baseline subtraction for stable small-batch optimization. Improved reasoning accuracy by up to 39.63 points over prior RL baselines and 67.97 points over the base model.

Rethinking the Design Space of Reinforcement Learning for Diffusion Models — **ICML 2026**

Jaemoo Choi*, Yuchen Zhu*, Wei Guo, Petr Molodyk, Bo Yuan, et al.

- Disentangled RL design into policy-gradient objectives, likelihood estimators, and rollout schemes, identifying ELBO-based final-sample likelihood estimation as the dominant factor. Improved SD3.5 Medium's GenEval score from 0.24 to 0.95 in 90 GPU hours, 4.6× more compute-efficiently than Flow-GRPO and 2× more efficiently than DiffusionNFT.

LaViDa-R1: Advancing Reasoning for Unified Multimodal Diffusion Language Models — **ICML 2026**

Shufan Li*, Yuchen Zhu*, Jiuxiang Gu, Kangning Liu, Zhe Lin, et al.

- Co-developed a unified SFT and multi-task RL post-training framework for multimodal diffusion LMs, using answer forcing, tree search, and complementary likelihood estimation to improve visual math reasoning, grounding, and image editing in a single model.
- Improved MathVista from 56.9 to 60.0, GSM8K from 47.4 to 81.5, and Lisa-Grounding P@0.5 from 29.2 to 66.7 over LaViDa-O, while raising ImgEdit from 3.71 to 3.90.